AGMIP7 Costa Rica

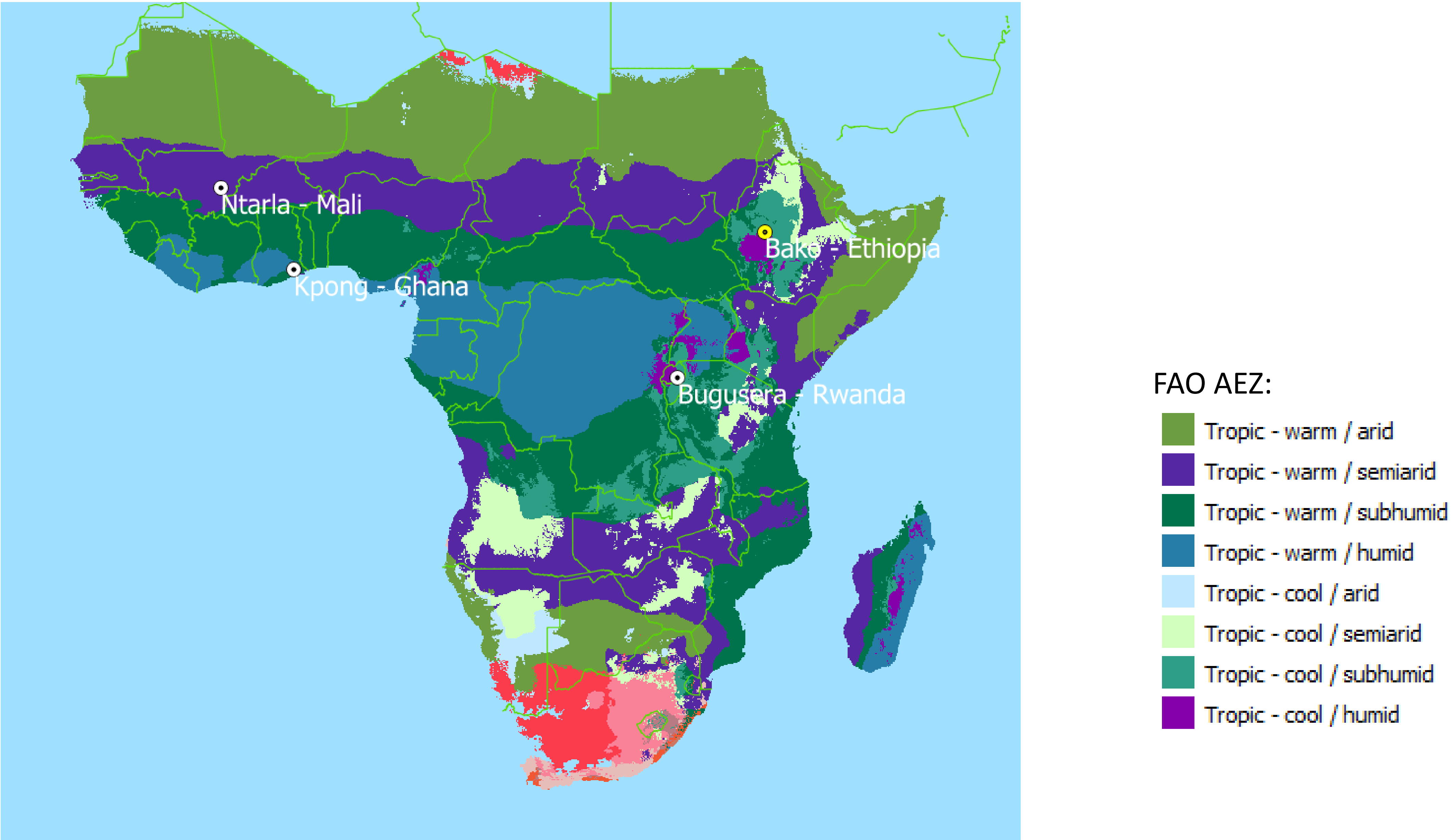
Crop Modeling of Low-Input Smallholder Systems

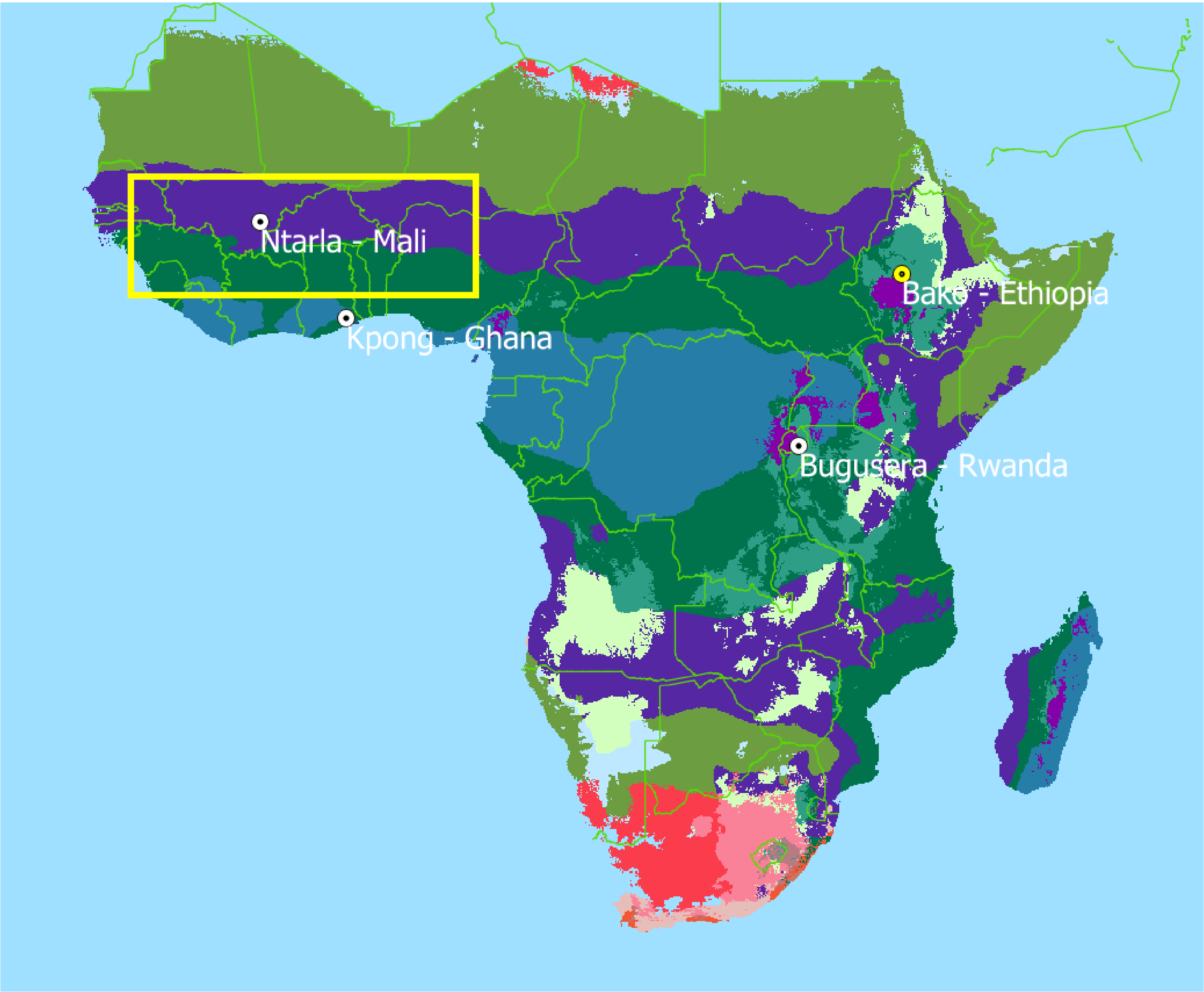
Available datasets from Sub Saharan Africa

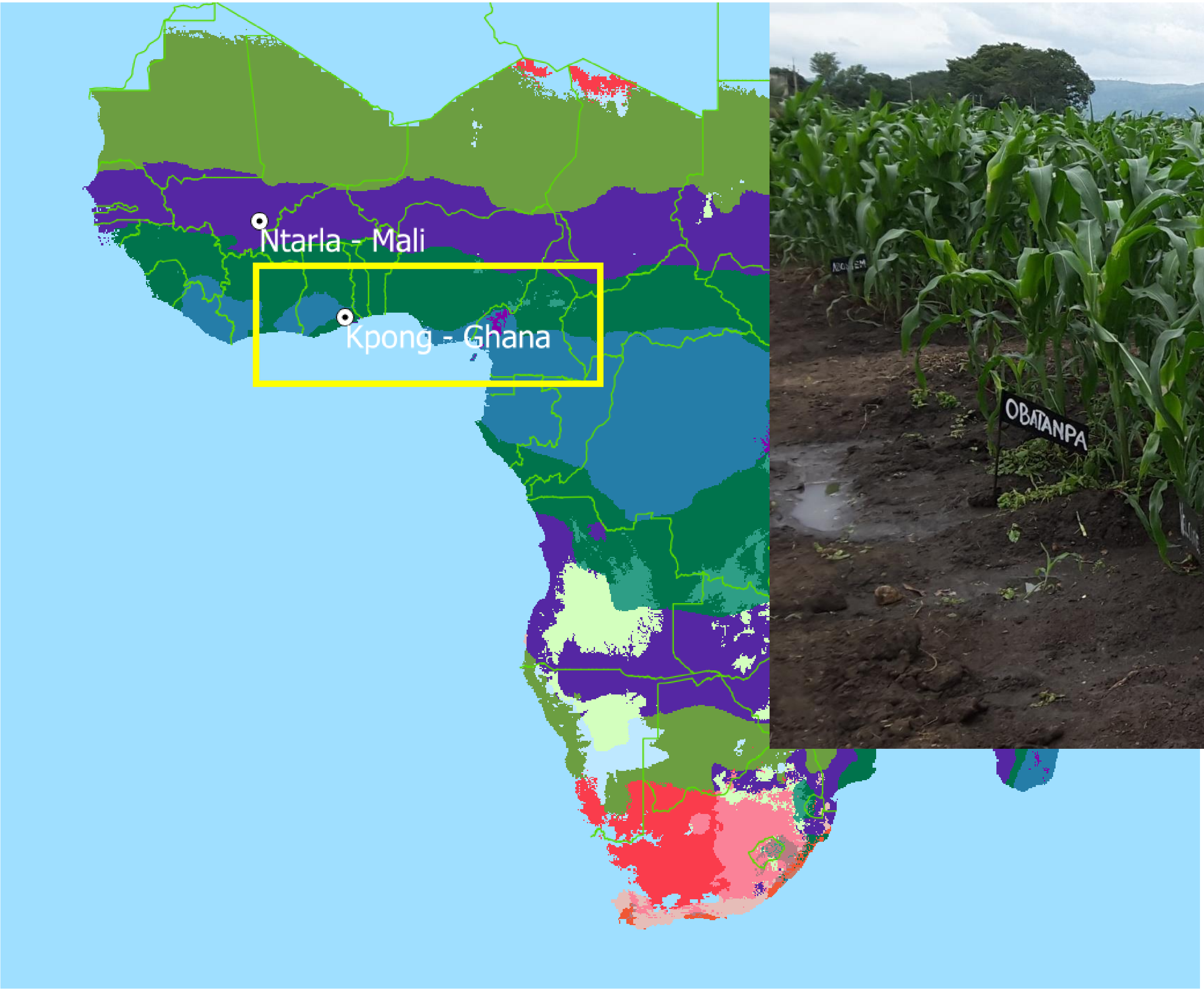
Co-organized by IICA

GLOBAL WORKSHOP • APRIL 24-26,2018 • SAN JOSÉ, COSTA RICA

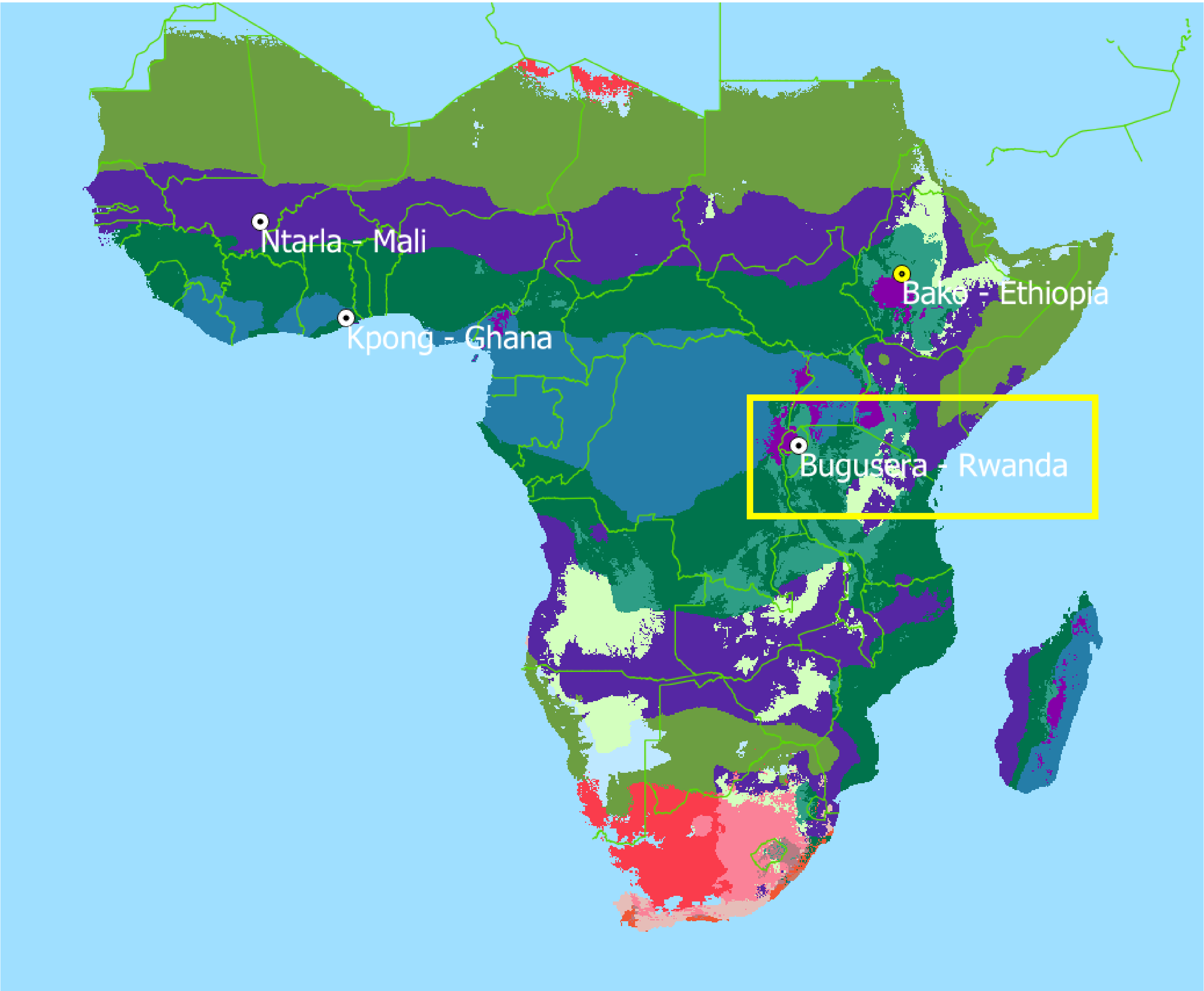
Four sites across sub-Saharan Africa

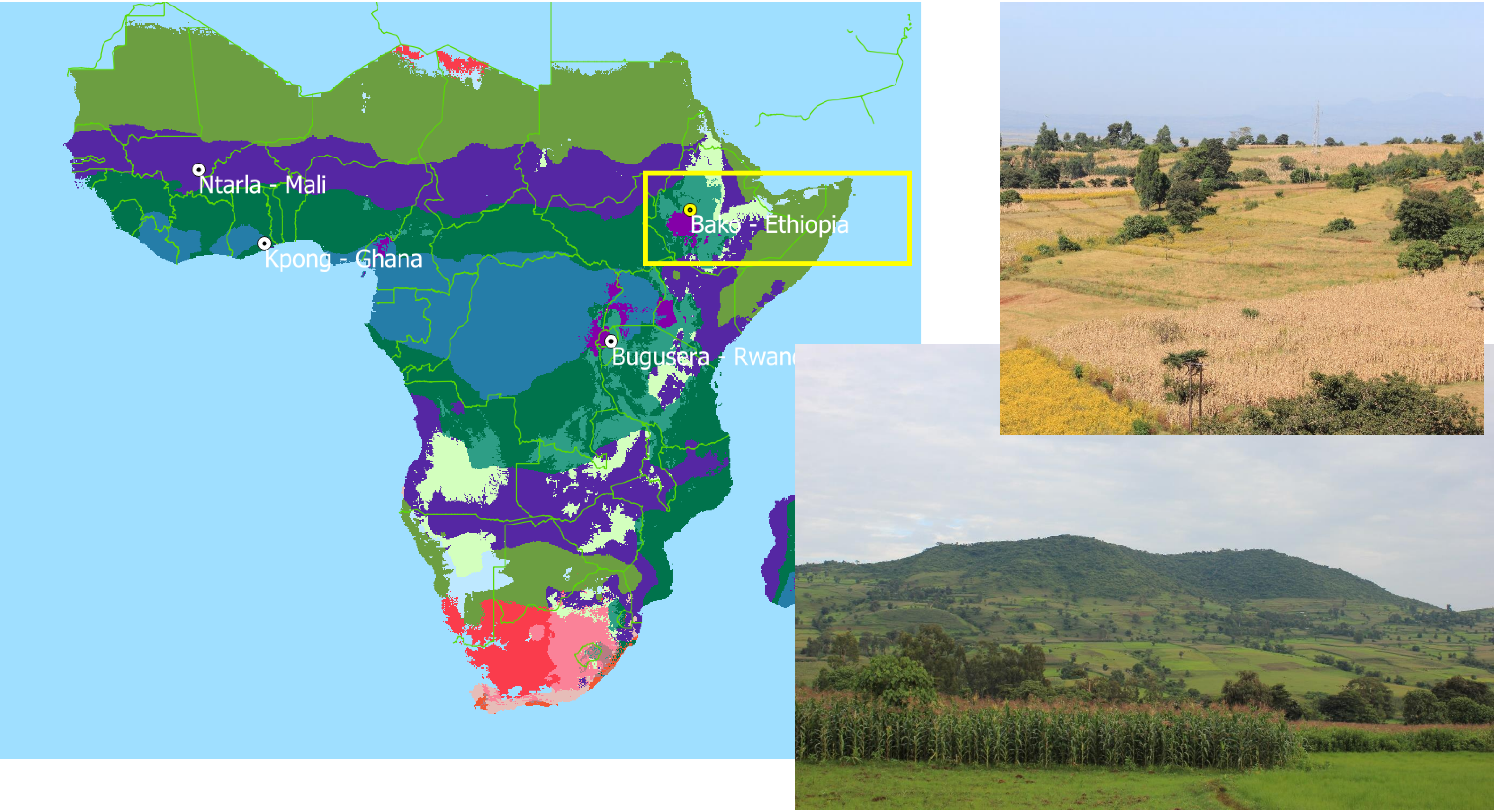



Ntarla - Mali
Kpong - Ghana
Bako - Ethiopia
Bugušera - Rwanda

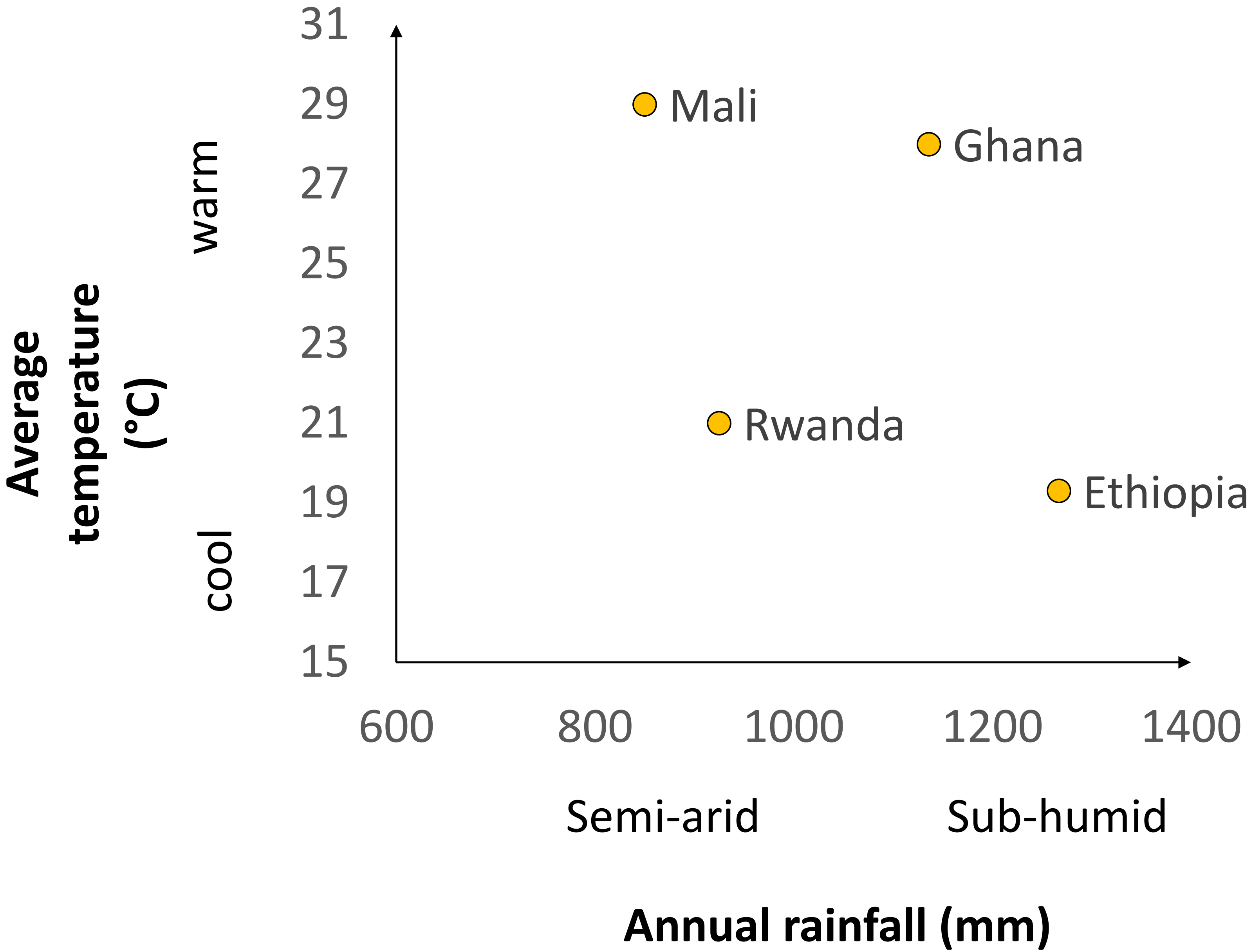


Ntarla - Mali
Kpong - Ghana


OBATANPA


Ntarla - Mali
Bako - Ethiopia
Kpong - Ghana
Bugusera - Rwanda


Ntarla - Mali
Kpong - Ghana
Bako - Ethiopia
Bugusera - Rwan

A diversity of Agro-ecological zones

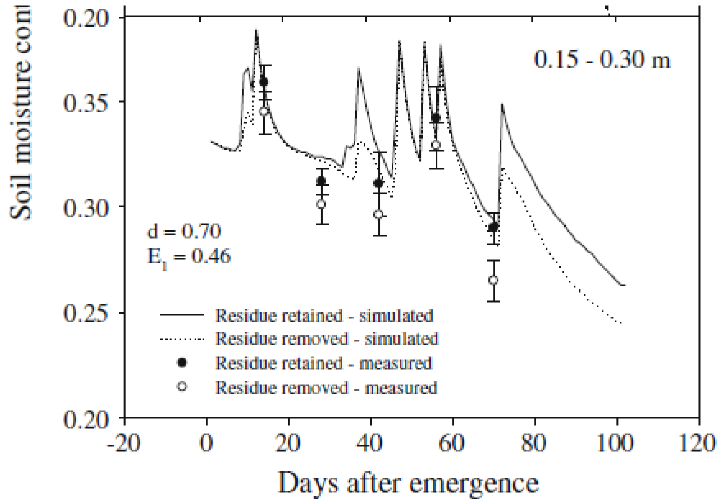


A diversity of soils under low input (rainfed, low N)

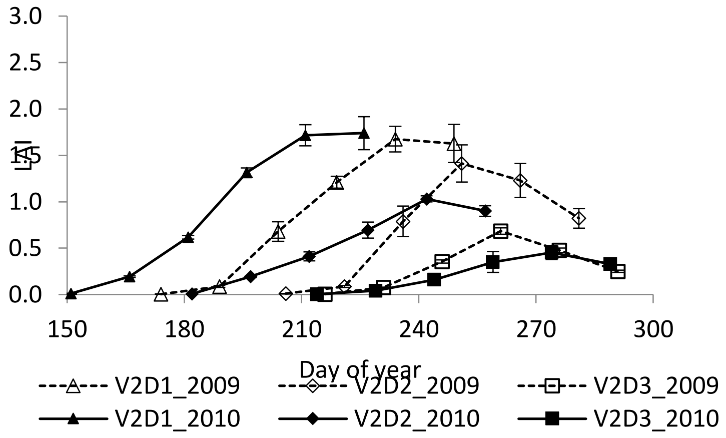
			MALI	GHANA	RWANDA	ETHIOPIA
			Ntarla	Kpong	Bugesera	Bako
SOIL	type		Ferric Lixisol	Vertisol	Humic Ferralsol	Nitisol
	texture		sandy	clayey	clayey	sandy
	pH		6	6	5.4	5.5
MANAGEMENT	cultivar		OPVs	Obatampa (OPV)	ZM607 (OPV)	BH540 (Hybrid)
	fertiliser treatments (kg/ha)	N	85	0, 40, 80, 120	64	86
		P	26	0,30	20	9
		K	16	0	0	0
	Irrigation		no	no	no	no
Average grain yield (t/ha)			2.1	3.3 (80N)	3.3	5.9

Measurements at all sites

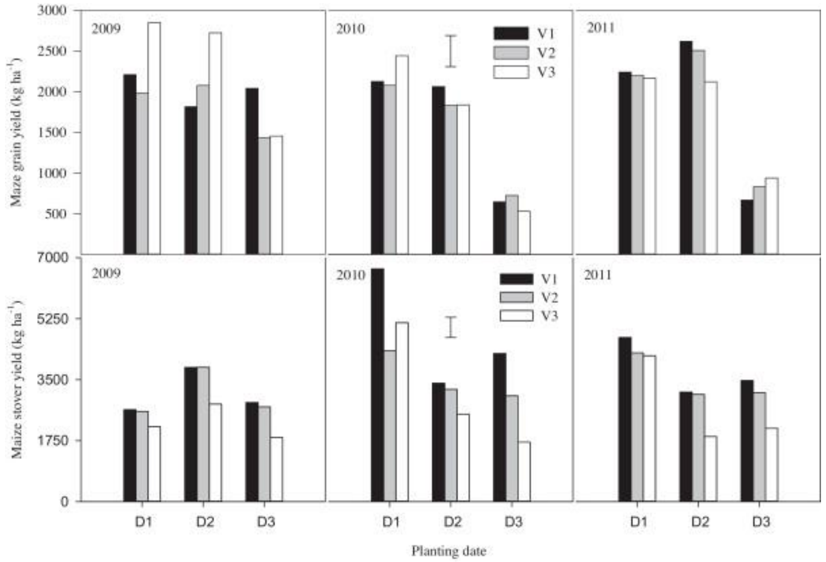
- Soil water dynamics



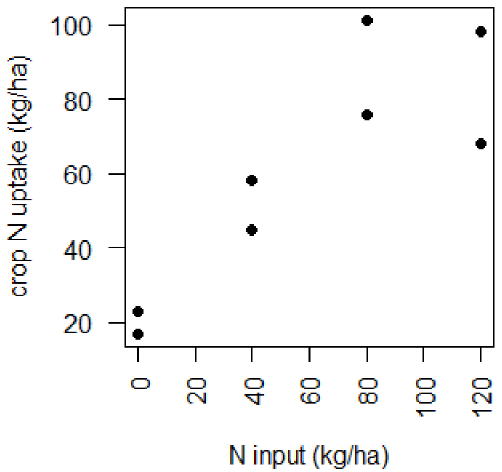
- LAI dynamics



- Total biomass, Grain yield



- Plant N (Ghana/Rwanda sites)



Weather

		MALI	GHANA	RWANDA	ETHIOPIA
		Ntarla	Kpong	Bugesera	Bako
Number of years		27	31	4	3
Period		1965-1993	1980-2010	2013-2016	2013-2015
Measurements	Min/Max temperature	x	x	x	x
	Rainfall	x	x	x	x
	Solar Radiation	x	x	x	x
	Wind speed			x	
	Relative humidity			x	

-> Nasa Power & Merra to estimate missing years (Rwanda/Ethiopia) and missing variables (Wind speed/Relative Humidity)

Overview of participating crop models

Crop model		Type of stress taken into account					CO_2 effect	Model type	Contact persons		
		Water	Nitrogen	Heat	Anoxia	Others					
1	APSIM	x	x	x	x		RUE/TE	Point model	Peter Thornburn	Enli Wang	Zhigan Zao
2	CERES-Maize	x	x	x	x		RUE/TE	Point model	Ken Boote	Cheryl Porter	
3	MAIZSIM	x	x	x	x		LF/TE/GY	Point model	Dennis Timlin	Soo-Hyung Kim	
4	Monica	x	x	x	x		RUE/TE/F	Point model	Claas Nendel		
5	SIMPLACE-Lintul	x	x	x	x		RUE/TE	Point model	Heidi Webber	Amit Srivastava	
6	MCWLA	x	x	x			F	Global model	Reimund Rötter	Tao Fulu	
7	PEGASUS	x	x	x			-	Global model	Delphine Deryng		
8	Info crop	x	x	x			RUE/TE	Point model	Naresh Kumar Soora		
9	RZWQM2	x	x	x			?	Point model	Laj Ahuja		
10	SALUS	x	x	x			RUE	Point model	Bruno Basso		
11	STICS	x	x	x			RUE/TE	Point model	Patrick Bertuzzi	Eric Justes	
12	Agro-IBIS	x	x	x			?	?	Tracy Twine		
13	HERMES	x	x		x		RUE/TE	Point model	Christian Kersebaum		
14	IXIM	x	x		x		RUE/TE	Point model	Jon Lisazo		
15	EPIC	x	x			x	RUE/TE	Global model	Curtis Jones		
16	GLAM	x		x			RUE/TE	Global model	Sinha Sumit	Marcelo Galdos	Andrew Challinor
17	CROPSYST	x	x	x			RUE/TE	Point model	Rolf Sommer	Sylvia Nyawira	
18	CELSIUS	x	x				RUE	Point model	François Affholder		
19	EXPERT-N-Ceres	x	x				RUE	Point model	Sebastian Gayler	Fasil Mequanint	
20	EXPERT-N-Sucros	x	x				RUE	Point model	Sebastian Gayler	Fasil Mequanint	
21	EXPERT-N-Spass	x	x				RUE	Point model	Eckart Priesack		
22	Hybrid-maize	x					-	Point model	Haishun Yang		
23	SARRA-H	x					?	Point model	Christian Barron		

Calibration procedures

- Low information phase :

  We give :

  - Soil characteristics (Texture/Field capacity/wilting point/ rooting depth)
  - Initial soil conditions (Water, N, SOC)
  - Sowing date, N input, amount and type of crop residues
  - Anthesis and maturity dates
  - Weather data

  Parameter adjustment : setting the time to anthesis and time to maturity observed in each site.

- High information procedure :

  We give :

  - dynamics of soil water, LAI, crop N
  - Total biomass/Yield

  Modeling groups adjust model parameters to improve the match with observed data, using your usual methods and documenting the changes.

Sensitivity to climate change parameters (T, CO2, rainfall)

Hypothesis to test:
Low soil fertility systems with minimal fertilizers may be less sensitive to weather and CO_2 changes, given other crop growth limiting factors.

Sampling design :

Objective	CO2 concentration (ppm)	Temperature change (°C)	Rainfall change (%)
Effect of temperature change	360	-3	0
		0	0
		3	0
		6	0
		9	0
Effect of CO2 concentration	360	0	0
	450	0	0
	540	0	0
	630	0	0
	720	0	0
Effect of rainfall change	360	0	-25
			-10
			10
			25
Combined effect of Temperature, CO2 and rainfall	540	3	-25
			25
		6	-25
			25
	720	3	-25
			25
		6	-25
			25

Achieved by modifying each day in the baseline climate series :

- Addition to minimum and maximum temperature
- Multiplicative factor for rainfall
- Imposed for CO_2 concentration

- **With fully calibrated models only, 22 simulations per site**

Comparing uncertainties of crops and global circulation models

Hypothesis to test : Uncertainty in predicting crop yields under climate change by current crop models is probably highest for low-input cropping systems of sub-Saharan Africa.

- 8 CMIP5 GCMS - Rcp 4.5/rcp 8. 5
  - "The eight GCMs were selected based on their climatic sensitivity, characterized by changes in annual mean temperature and precipitation, to preserve the range of uncertainties in CMIP5" (Tao et al. 2017)"
  - Temperature/rainfall changes imposed on historical series following delta approach
- Three levels of intensification (N inputs) : 0-50-100 kg/ha
- Two sowing dates
- 8*2*3*2 = 96 simulations per site… !

With fully calibrated models only

Templates / data sharing

Inputs

	A	B	C	D	E	F	G
1	**Tansania**	**Low information**					
2							
3	**Year**	2009 sown and 2010 harvested					
4	**Location**	Morogoro					
5	**Name weather station**	Morogoro					
6	**X (°)**	S 06° 50'					
7	**Y (°)**	E 37° 39'					
8	**Altitude**	500					
9	**Soil type**	**Ultic Haplustalfs**					
10							
11	**Clock of the simulation**	Starting on*	290	*only for low	input phase		
12							
13							
14							
15							
16							

General information | Baseline | A2_Future scenario | Soil permanent characteristics

Soil initial cond_low inform | Cultivar | Management | ALL_INFORMATION

Dropbox

On dropbox ?

Templates / data sharing

Outputs

	Parameter:	Unit:
5	**Parameter:**	**Unit:**
6	Site/country	
7	Latitude	
8	Longitude	
9	Altitude	masl
10	Year	
11	Crop	
12	Date of sowing	Date
13	Date of emergence (Zadocs 12)	Date
14	Date of start of flowering (Zadocs 51)	Date
15	Date of start of anthesis (Zadocs 60)	Date
16	Date of physiological maturity (Zadocs 90)	Date
18	Final grain yield (dry matter in grains) or dry weight of storage organs	kg ha^{-1}
19	(Maximum) above-ground biomass (dry weig	kg ha^{-1}
20	(Maximum) below-ground biomass (dry weig	kg ha^{-1}
21	(Maximum) above-ground N-uptake	kg N ha^{-1}

Basics Dynamics

2	**Model intercomparison**			
3	**Output format for model results**			
4				
5	Site/country	*(The idea is to make this kind of sheets for each studie*		
6	Year			
7				
8	**Day in Julian calendar (1st Jan = 1)**	**Soil moisture content 0-90 cm or profile depth [volume fraction]**	**Actual evapo-transpiration [mm day^{-1}]**	**Green leaf area index [-]**
9	1			
10	2			
11	3			
12	4			
13	5			
14	6			
15	7			
16	8			
17	9			
18	10			
19	11			

Basics Dynamics

-> The idea is to make this kind of files (two sheets) for each site and year

Calendar

		2018							
	A	M	J	J	A	S	O	N	D
Protocol development									
Protocol discussion with modelling groups									
Low information calibration and sensitivity analysis CO2, T, Rainfall, N									
High information calibration and sensitivity analysis CO2, T, Rainfall, N									